Memory Mapping Experiment

Methods Documentation

Noa Herz, Nov 7, 2023

Version	Authors	Description	Date
1.0	N. Herz	Initial release	

Study Overview

Stimulation will be delivered to map brain regions that are crucial for verbal memory function.
In this experiment, participants will be presented with (non-semantically associated) pairs of words. They will then be required to complete: 1. cued-recall phase, 2. Recognition (old/new) phase. This design will be used as a provocation test for the prediction of verbal memory decline after hippocampectomy in epilepsy patients.

General Design

- Instructional video preceding the execution of the task
- 3 word-pairs per list; 12 lists per session
- 10s countdown before the encoding phase of each list
- Encoding (~18s): 250 ms beep + 500-750 ms jittered blank + 4000 ms word pair + 1000 ms ISI
- Distractor (>=20s): A+B+C = ? math problems, where A/B/C are random integers 1-9
- Cued recall (~27s): 250 ms beep + 500-750 ms jittered blank + 4000 ms word cue + 4000 ms post-cue
- Recognition (~42 s): 250 ms beep + 500-750 ms jittered blank + 4000 ms word probe + 2000 ms post-cue

Time per list: >=117 s. Time per 12 lists: ~24 minutes

Initial mapping stage:

This stage will be carried out to determine which brain region is the most likely to function as a memory eloquent area. Stimulation location will change each trial (stim locations across the MTL will be pre-determined; see priorities below). The stimulated region leading to the greatest memory reduction will be defined as a memory node for subsequent testing.

In this short session, participants will complete one of every list condition (see three conditions below: no-stim list, encoding stim, retrieval stim list) in random order. The list condition should be visible only to the experimenter while they are running the study (through the BlackRock PC). This procedure will be carried out for each pre-determined stim site. Lists will include two pairs of items (instead of 3 as in the full design). The rest of the task characteristics are the same as in the full design.

Measures extracted in the initial mapping stage:

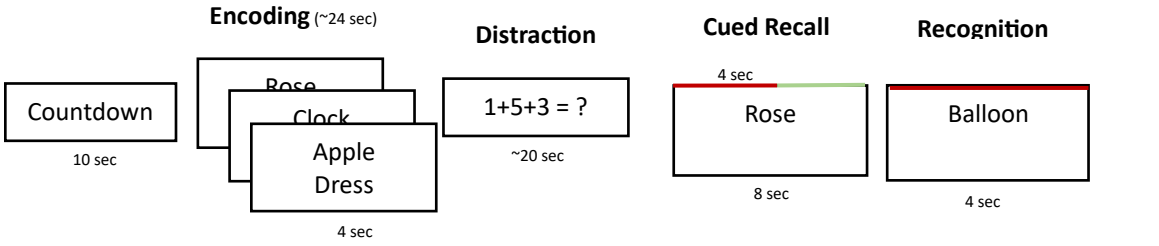
Memory hubness will be measured as:

1. Encoding: The number of words recalled from the stim vs. no-stim lists in either recognition or cued-recall. A site showing the greatest decline will be regarded as a memory formation node.
2. Retrieval: the number of words correctly remembered in the cued-recall/recognition in the retrieval stim vs. no-stim lists. The number of words recalled in the cued recall in the initial retrieval time (those remembered during the 4 initial seconds of recall phase vs. the 4-second non-stim phase of each trial) will also be taken into account.

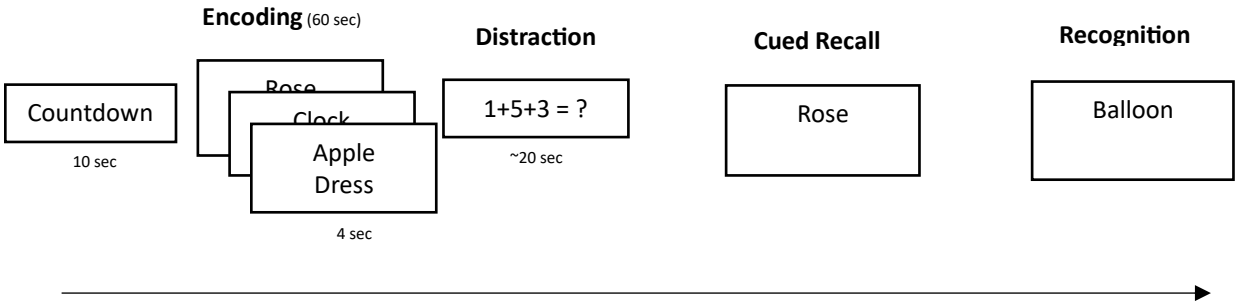
1. No-stim list: No stimulation is delivered in any of the task phases.
2. Encoding stim lists: Stimulation will be delivered throughout the presentation of item pairs (4 sec) at encoding to investigate whether the stimulated area is crucial for memory formation.
3. Retrieval stim lists: Stimulation will be delivered during the cued-recall and recognition phases to test whether the retrieval of items is disrupted by stimulation.

Full task (after choosing the stim site):
Each session will include 12 lists, randomly chosen in batches of three, one from each condition: encoding stim list (4 lists), no-stim list (4 lists), and retrieval stim list (4 lists). Each list will consist of 3 trials (pairs of words).
During the cued-recall phase, patients will be required to recall items vocally. Item recall will be annotated offline. Responses in the memory recognition will be done using a keypress.

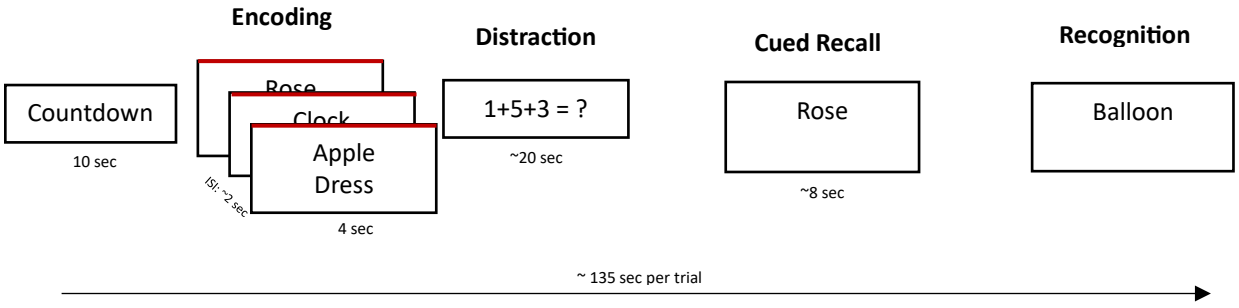
Task Schematic:
Retrieval stim list:



No-stim list:



Encoding stim list:



A question about awareness of stimulation will be delivered at the end of the experiment to ensure that subjects were blind to the stim condition (see below).

Stimulation Parameters (should be set in a way that enables changing them, if needed)**:**
Biphasic rectangular pulses, pulse width: 600 microsecond pulse (300 us per phase)

Stim site: MTL: entorhinal > hippocampus > posterior parahippocampal gyrus > amygdala (left hemisphere will always be prioritized unless clear indication exists for a right-hemisphere dominance). Most patients have three contacts in the hippocampus and one in the amygdala. Each session will test a different stim site.
Memory phase: encoding/retrieval
Stim frequency: 50 Hz
Stim amplitude: 0.7 mA (or below the after-discharge threshold).
Stim duration: 4.1 sec during item presentation (either during item presentation at encoding / during the presentation of the cue word at the cued-recall phase / during the presentation of the recognition probe)

Specific methods:

The task is divided into a set of lists, which include a study phase and two test phases each.

Study phase: participants sequentially see three trials of word pairs (AB) and are instructed to remember the arbitrary associations between them. Word pairs on each list will be randomly sampled from the pool of 300 common nouns (23,24) (and thus, word pairs will not be identical across participants). Words will not repeat across lists or across the first two sessions of each subject (all words are sampled without repetition). Each word pair is preceded by a short fixation period (250–300 ms) with a low beep, followed by a blank inter-stimulus interval (ISI) between 500–750 ms. Word pairs are presented stacked in the center of the screen (one above the other) for 4000 ms, followed by a blank ISI of 1000 ms.

- On encoding stim lists, stimulation will begin -50 ms prior to item presentation, with a jitter of +-50 ms (hence, stim will start from 100 ms prior to 0 ms prior to item presentation). Stimulation will last 4.1 sec. All pairs are stimulated.

Distraction: To interrupt participants' active maintenance of the word pairs prior to the test phase, participants are probed to complete an arithmetic distractor task of the form 'A + B + C = ?' for a minimum of 20 seconds after the study phase. The distractor phase ends after a response is submitted (by typing). ABC are integers from 1-9.

Cued recall: The appearance of a row of asterisks, along with an audible tone, signals the start of the cued recall phase. In this test phase, participants are cued with one word from each encoded pair, selected in a random order. Subjects are given a cued-recall test on all of the word pairs in the study list. Some of the cues are in the forward direction (A-?), and some are in the backward direction (?-B). The order (e.g., forward, backward, backward) will be randomly chosen from four random options. The chosen order will then be matched between the three stim conditions (no-stim, encoding stim, retrieval stim) and then shuffled across the four batches. That way, subjects will not be able to predict the order of trials in the subsequent trial, but the order will be similar across the three stim conditions. Cues are presented in the middle of the screen. Each cue is preceded by a low beep, followed by a blank inter-stimulus interval (ISI) between 500–750 ms. Each cue word is presented for 4000 ms followed by a blank ISI (black screen) of 4000 ms, making a total of 8000 ms as the maximum recall time window. Participants are instructed to say the associated word aloud, with their responses recorded by a microphone. Participants can vocalize their responses at any time during the recall period after the cue presentation. They are instructed to guess if they can't retrieve the target word. After 4 seconds, a key (space bar) enables participants to move on to the next trial. Without pressing the key, the trial ends 8 sec from the beginning of the trial. The text: "(please speak now)" appears at the center lower part of the screen throughout the cued-recall phase.

- On retrieval stim lists, stimulation will begin -50 ms prior to cue presentation, with a jitter of +-50 ms (hence, stim will start from 100ms prior to 0 ms prior to cue presentation). Stimulation will last 4.1 sec. All cues are stimulated.

Recognition: The appearance of an old/new instructions slide signals the beginning of the recognition test. Patients are asked to press a key when ready to begin this phase. Subjects are given a recognition test on all of the trials in the study list. If the B member of a pair was presented in the cued recall, recognition is tested on the A item; if the A member of a pair was tested in the cued recall, recognition is tested on the B item. In this way, each word is always presented only once during the test phase.
Patients are asked to make old-new judgments by pressing one of two buttons on a computer keyboard with their right or left index finger. The specific keys used for indicating target and lure responses ("old"/"new") are counterbalanced across subjects. The text: "Old" and "New" appear at the correct screen location (center right/left) prior to the beginning of the recognition phase as a reminder for participants (immediately after the instructions slide).
Probes are presented in the middle of the screen. Each probe is preceded by a low beep, followed by a blank inter-stimulus interval (ISI) between 500–750 ms. Patients are given a maximum of ~6 sec to respond to each probe. Probe items are presented for 4000 msec, followed by 2000 ms of blank ISI. Then, the next probe pair is presented. After 4 seconds, a key (space bar) enables participants to move on to the next trial (otherwise, the next trial will begin automatically after 2 sec).
A total of 6 trials are included in the recognition test; half (i.e., 3) are previously studied, old items; the remainder (3 items) are new (the order of recognition trials is randomly chosen). The three new item pairs are randomly selected from the same word pool without repetition. Both response accuracy and RT are recorded in this task (all responses are recorded, even if more than one response is made).

- On retrieval stim lists, stimulation will begin -50 ms prior to probe presentation, with a jitter of +-50 ms (hence, stim will start from 100ms prior to 0 ms prior to probe presentation). Stimulation will last 4.1 sec. All probes are stimulated.

The order of the six items in the recognition phase is randomly assigned across the four lists included in each stim-condition (let's call them orders 1-4). The chosen order will be used in each of the three list conditions, though in a randomly assigned order (for example, the no-stim lists will use order 4,2,3,1, the encoding stim lists will use order 3,1,4,2, and the recall-stim lists will have order 2,1,4,3). In that way, each list condition will include a similar order overall without allowing patients to recognize the order.

Stimulation awareness:
Each session will end with the question:
"Can you recall any specific moments during the experiment when you knew or felt stimulation was being delivered?"

Yes / No

(If yes):
"Please describe when and why you think stimulation was delivered."
(Please speak now. Press any key when finished).

Recognition instructions:
We will now show you several words. Some of the words were part of the word pairs shown in the previous list.
For each word, please indicate whether it was part of the presented pairs (X key) or not (Y key).

Memory Deterioration Following Surgery
Neuropsychological examination for memory function will be conducted before and after surgery, with follow-up at 1 month and 1 year to determine if reorganization compensated for initial verbal memory loss. WMS-R verbal memory, visual memory, general memory, and delayed recall scores will be extracted.

Main Questions:

- Memory hub was resected: Did memory improve/decline following surgery?
- The hippocampus was resected, but no memory hubness was identified (if stimulation did not reduce memory in any location)

- Memory hub was not resected: Did memory improve/decline following surgery?

Relevant References:
https://thejns.org/view/journals/j-neurosurg/125/5/article-p1053.xml?tab_body=fulltext
https://academic.oup.com/brain/article/128/11/2742/339579

Future directions:

- Long-term memory (testing memory over days instead of on the same day): https://www.sciencedirect.com/science/article/pii/S0035378717305192?casa_token=7jeumJ-UbGYAAAAA:5rthH0WrFqr0-2u-XjB-ojLbMMqzGwJanstnXgieSWfohw8EkQLaLuGsPVp3bcL_zdcKZIg
- Theta phase during the presentation of stimulus may influence how words are remembered. Consider controlling the presentation of items relative to that.

2 do

- Add a practice list – the same as in PAL.
- Add a low beep to orient patients to the subsequent stimuli
- Find out whether 300us per phase was used in the stim experiments at Penn and ensure safe stimulation parameters.